

Formalizing building-up construction of self-dual codes through isotropic lines in Lean

Jae-Hyun Baek

Department of Artificial Intelligence and
Institute for Mathematical and Data Sciences
Sogang University, Seoul, Korea

Jon-Lark Kim

Department of Mathematics and
Institute for Mathematical and Data Sciences
Sogang University, Seoul, Korea

September 5, 2026

Abstract

The purpose of this paper is two-fold. First, we show that, after a specified form isometry, the two-coordinate reduction in the binary Hilbert-symbol realization of Chinburg and Zhang is inverse to Kim's building-up construction, up to permutation equivalence. Second, for $q \equiv 1 \pmod{4}$, we develop a q -ary analogue of this reduction-and-extension mechanism. The identity $c^2 = -1$ yields the isotropic line governing the split construction. For every fixed ordered pairing of the coordinates, we obtain a universal rank- r boxed normal form, where r is the dimension of the intersection with the product of these isotropic lines; after a scalar-coordinate permutation, every nonzero-length Euclidean self-dual code has a rank-one representative. Applications include a recursive optimal pair over $\text{GF}(5)$ and an exact repeated boxed realization of a self-dual $[18, 9, 8]$ code over $\text{GF}(13)$ and a self-dual $[20, 10, 10]$ code over $\text{GF}(13)$, in which the rank-one parent and its building-up child occur in one complete generator matrix. The algebraic core is formalized in Lean 4.

1 Introduction

Coding theory is the study of reliable communication. Its modern foundations were laid by Shannon's seminal paper [19]. Since then, many classes of linear codes have been studied. In particular, the extended binary Hamming code of length 8, the binary Golay code of

length 24, and the ternary Golay code of length 12 are self-dual, that is, $\mathcal{C} = \mathcal{C}^\perp$. Thus the classification or construction of self-dual codes has been one of the most active research problems [17]. Self-dual codes have wide connections with several mathematical areas including t -design theory [1], group theory [5], number theory [2], [5], [20]. Furthermore, they have applications in quantum error-correcting codes [3].

Kim [8] introduced “building-up construction” which proved that all binary self-dual codes can be constructed in this way. This construction starts from a binary self-dual code of length $2n$ and then adds a vector of length $2n$ of odd weight to construct a binary self-dual code of length $2n + 2$. Kim–Lee [11], [10] generalized this method to construct self-dual codes over finite fields $\mathbb{F}_q$, where q is even or $q \equiv 1 \pmod{4}$. In practice, this is a principal method for constructing examples from a split bilinear space. However, the formula for the new generator rows is often presented in terms of an auxiliary vector whose inner product with itself is -1 . One of the central points of the present paper is that the corresponding correction coordinates are naturally organized by an isotropic line in a hyperbolic plane, and that the forward construction admits a precise row-wise and family-wise characterization in those terms.

On the other hand, Kreck and Puppe [12] constructed binary self-dual codes from the cohomology of three-manifolds. Chinburg and Zhang [4] developed the arithmetic analogue for rings of S -integers. For a suitable set S of places of a number field, the image of the global-to-local map in first étale cohomology is Lagrangian for the sum of the local Hilbert pairings. Their Theorem 1.5 shows that, for $K = \mathbb{Q}$, every binary self-dual code of length at least four occurs up to permutation equivalence. We use precisely this realization theorem and its boxed reduction; no further étale-cohomological machinery is needed below.

In this paper, we focus on $q \equiv 1 \pmod{4}$. Over $\mathbb{F}_q$, self-dual codes are maximal totally isotropic subspaces for the standard Euclidean bilinear form on $\mathbb{F}_q^{2n}$. In this split case the elementary identity $c^2 = -1$ is the basic algebraic input behind both the classical building-up construction and the structural description of the ambient quadratic space: it produces an isotropic line in $\mathbb{F}_q^2$, identifies the Euclidean plane with a hyperbolic plane, and makes the norm form $x^2 + y^2$ split.

We establish that each generator row of the building-up admits the isotropic line decomposition $r_i = -y_i \cdot (1, c, 0, \dots, 0) + (0, 0, g_i)$, where $(1, c)$ spans an isotropic line in the hyperbolic plane. This decomposition provides a precise geometric interpretation of the correction terms in the building-up construction. We also give a constructive norm-form: once $c^2 = -1$ and $2 \neq 0$, every $a \in \mathbb{F}_q$ can be written explicitly as $a = x^2 + y^2$, which in turn yields extension vectors of prescribed norm.

The odd-characteristic and binary inputs are distinct. For $q \equiv 1 \pmod{4}$, the element -1 is a square in $\mathbb{F}_q^\times$, so the Euclidean plane is hyperbolic: the vectors

$$h_1 = (1, c), \quad h_2 = \frac{1}{2}(1, -c)$$

satisfy $h_1 \cdot h_1 = h_2 \cdot h_2 = 0$ and $h_1 \cdot h_2 = 1$. In the binary theorem of Chinburg–Zhang [4], the arithmetic image is Lagrangian for a cohomological pairing; selected orthonormal coordinates

identify that pairing with the Euclidean dot product. The coordinate plane deleted in the resulting binary reduction has Gram matrix I_2 and is non-alternating.

In the present paper, this distinction organizes the binary comparison, the split q -ary construction, its universal normal form, and selected applications over $\text{GF}(5)$ and $\text{GF}(13)$.

Our main contributions are as follows.

- (1) In the binary case, we distinguish the cohomological pairing from the Euclidean dot product by a specified form isometry. In the resulting Euclidean coordinates, boxed reduction deletes a two-coordinate block with Gram matrix I_2 , the tail rows form a self-dual predecessor, and the remaining coefficients satisfy Kim’s correction formula (Theorem 3.4).
- (2) In the q -ary case, we prove the adapted reduction and extension theorem, a forward split boxed construction, and its reverse statement under an explicit adapted-presentation hypothesis (Theorems 3.9–3.13).
- (3) For each fixed ordered coordinate pairing, every Euclidean self-dual code over $\mathbb{F}_q$, $q \equiv 1 \pmod{4}$, has the universal rank- r boxed normal form of Theorem 3.15, with $r = \dim(\mathcal{C} \cap U_c)$. After a scalar-coordinate permutation, one may always take $r = 1$, as proved in Theorem 3.16.
- (4) The applications consist of two optimal self-dual codes over $\text{GF}(5)$ related by a boxed reduction and extension, and one complete repeated boxed realization of a self-dual $[18, 9, 8]$ code over $\text{GF}(13)$ and a self-dual $[20, 10, 10]$ code over $\text{GF}(13)$, displayed as an exact parent–child generator (Section 4).
- (5) The modular Lean 4 development formalizes the paper-facing algebraic interfaces without `sorry` or `native_decide`; independent Challenge/Solution comparisons check the exact theorem statements.

The cited arithmetic and coding-theoretic realization theorems remain mathematical inputs. Lean verifies the finite-dimensional algebra and the displayed construction, reduction, and normal-form interfaces; the distance enumerations are independent exact computations.

The paper is organized as follows. Section 2 fixes notation. Section 3 proceeds from the binary case to the split q -ary case and finally to the universal normal form. Section 4 contains the selected applications, and Section 5 concludes.

2 Preliminaries

We refer to [7] for basic concepts and facts and to [17] for self-dual codes.

Throughout the paper, $\mathbb{F}_q$ denotes a finite field. We equip $\mathbb{F}_q^n$ with the Euclidean bilinear form

$$\langle u, v \rangle := \sum_{i=1}^n u_i v_i.$$

For a linear code, the minimum distance equals the minimum Hamming weight of its nonzero codewords. For a linear code $\mathcal{C} \subseteq \mathbb{F}_q^n$, we write

$$\mathcal{C}^\perp := \{v \in \mathbb{F}_q^n : \langle v, c \rangle = 0 \text{ for all } c \in \mathcal{C}\}.$$

For subspaces $\mathcal{A}, \mathcal{B} \subseteq V$, put $\mathcal{A} \cap \mathcal{B} := \{v \in V : v \in \mathcal{A} \text{ and } v \in \mathcal{B}\}$. For a matrix G over K , $\langle G \rangle_K$ denotes the code spanned by its rows.

Definition 2.1 (self-dual code) A linear code $\mathcal{C} \subseteq \mathbb{F}_q^n$ is *self-dual* if $\mathcal{C} = \mathcal{C}^\perp$ [16].

Definition 2.2 (Lagrangian subspace) Let V be a finite-dimensional vector space over $\mathbb{F}_q$, equipped with a nondegenerate symmetric bilinear form. A subspace $L \subseteq V$ is called *Lagrangian* [14, 18] if $L = L^\perp$.

For a nondegenerate form, $\dim L^\perp = \dim V - \dim L$. Hence $L = L^\perp$ if and only if L is totally isotropic and $2 \dim L = \dim V$. In particular, a bilinear space admitting a Lagrangian subspace is even-dimensional. Applied to the standard Euclidean form, this says that $\mathcal{C} \subseteq \mathbb{F}_q^n$ is self-dual if and only if it is totally isotropic and $2 \dim \mathcal{C} = n$. Thus a self-dual code is precisely a Lagrangian subspace of $(\mathbb{F}_q^n, \langle \cdot, \cdot \rangle)$, and its length is even.

Listing 1: Lean for Definitions 2.1 and 2.2.

```
def paperSelfDualCode (C : Submodule K (Fin n → K)) : Prop :=
  C = (dotBilin (K := K) (n := n)).orthogonal C

def paperLagrangianSubspace (L : Submodule K (Fin n → K)) : Prop :=
  L = (dotBilin (K := K) (n := n)).orthogonal L

theorem paper_self_dual_iff_lagrangian :
  paperSelfDualCode (K := K) C ↔ paperLagrangianSubspace (K := K) C

theorem paper_dotBilin_nondegenerate :
  (dotBilin (K := K) (n := n)).Nondegenerate

theorem paperLagrangianSubspace_iff_totallyIsotropic_and_finrank_half :
  paperLagrangianSubspace (K := K) L ↔
    IsTotallyIsotropic (dotBilin (K := K) (n := n)) L ∧
    2 * Module.finrank K L = Module.finrank K (Fin n → K)

theorem paperSelfDualCode_iff_totallyIsotropic_and_finrank_half :
  paperSelfDualCode (K := K) C ↔
    IsTotallyIsotropic (dotBilin (K := K) (n := n)) C ∧
    2 * Module.finrank K C = Module.finrank K (Fin n → K)
```

```

theorem paperLagrangianSubspace_even_length
  (hL : paperLagrangianSubspace (K := K) L) : Even n

theorem paperSelfDualCode_even_length
  (hC : paperSelfDualCode (K := K) C) : Even n

```

Proposition 2.1 (Systematic form criterion) *Let $\mathcal{C} \subseteq \mathbb{F}_q^{2k}$ be generated by a matrix $G = [I_k \mid A]$ with $A \in M_k(\mathbb{F}_q)$. Then $\mathcal{C}$ is self-dual if and only if*

$$AA^T = -I_k.$$

Proof. The rows of G span a k -dimensional code. Hence self-duality is equivalent to pairwise orthogonality of the rows. The Gram matrix GG^T of the rows is

$$GG^T = I_k + AA^T = O,$$

so the orthogonality condition is exactly $AA^T = -I_k$. □

Listing 2: Lean for Proposition 2.1.

```

theorem paper_systematic_form_criterion_exact
  (A : Matrix (Fin k) (Fin k) K) :
  paperSelfDualCode (K := K) (rowSpace (systematicRows A)) ↔
  A * A.transpose = -(1 : Matrix (Fin k) (Fin k) K)

```

Definition 2.3 (Permutation equivalence) Writing codewords as row vectors, two codes $\mathcal{C}_1, \mathcal{C}_2 \subseteq \mathbb{F}_q^n$ are *permutation equivalent* if there is a permutation $\sigma \in S_n$ such that

$$\mathcal{C}_2 = \mathcal{C}_1 P_\sigma,$$

where P_σ is the permutation matrix associated with σ .

In the Lean notation below, `permuteVec  $\sigma$  u` denotes the row-vector product uP_σ . Coordinate permutations are Euclidean isometries:

$$\langle uP_\sigma, vP_\sigma \rangle = \langle u, v \rangle.$$

Consequently, permutation equivalence preserves Euclidean self-duality. A general monomial transformation also permits independent nonzero coordinate scalings; it need not preserve the Euclidean form or self-duality. It is therefore not included in our use of “equivalence” below. Over $\mathbb{F}_2$, permutation and monomial equivalence coincide because $\mathbb{F}_2^\times = \{1\}$.

Listing 3: Lean for Definition 2.3.

```

def CodeEquiv (G1 : Fin 1m → Fin n → K)

```

```

(G2 : Fin 2m → Fin n → K) : Prop :=
  ∃ σ : Equiv.Perm (Fin n),
    rowSpace (permuteFamily σ G1) = rowSpace G2

theorem dot_coordinatePermuteLinearEquiv :
  dot (permuteVec σ u) (permuteVec σ v) = dot u v

theorem codeEquiv_preserves_paperSelfDualCode
  (hG1G2 : CodeEquiv G1 G2)
  (hG1 : paperSelfDualCode (K := K) (rowSpace G1)) :
  paperSelfDualCode (K := K) (rowSpace G2)

```

From this point onward, when discussing the split q -ary construction we assume that q is odd and satisfies $q \equiv 1 \pmod{4}$. The condition $q \equiv 1 \pmod{4}$ is equivalent to -1 being a square in $\mathbb{F}_q^\times$. We fix once and for all an element

$$c \in \mathbb{F}_q^\times \quad \text{with} \quad c^2 = -1.$$

This choice is used throughout the building-up formulas.

Definition 2.4 (hyperbolic plane) A hyperbolic plane [18] over $\mathbb{F}_q$ is a 2-dimensional symmetric bilinear space with basis $\{e_1, e_2\}$ satisfying

$$\langle e_1, e_1 \rangle = \langle e_2, e_2 \rangle = 0, \quad \langle e_1, e_2 \rangle = 1.$$

Such a basis $\{e_1, e_2\}$ is called a hyperbolic pair. Any 1-dimensional totally isotropic subspace is called an isotropic line.

In even dimension and odd characteristic, a nondegenerate symmetric bilinear space is called *split* if it is an orthogonal direct sum of hyperbolic planes. Thus splitting is a property of the bilinear space, not of the underlying finite field.

Proposition 2.2 *Let q be odd and let $c \in \mathbb{F}_q$ satisfy $c^2 = -1$. Then:*

- (i) *The element c has multiplicative order 4.*
- (ii) *For every $k \geq 1$, the matrix equation $AA^T = -I_k$ has the explicit solution $A = cI_k$.*
- (iii) *The binary quadratic form $x^2 + y^2$ splits as*

$$x^2 + y^2 = (x - cy)(x + cy),$$

hence it represents every element of $\mathbb{F}_q$.

Proof. For (i), since q is odd, $-1 \neq 1$. Thus $c^4 = 1$ and $c^2 \neq 1$, so c has order 4.

For (ii), take $A = cI_k$. Then $AA^T = c^2I_k = -I_k$.

For (iii),

$$(x - cy)(x + cy) = x^2 - c^2y^2 = x^2 + y^2.$$

Given $a \in \mathbb{F}_q$, take

$$x = \frac{1+a}{2}, \quad y = \frac{1-a}{2}c.$$

Then $x - cy = 1$, $x + cy = a$, and consequently $x^2 + y^2 = a$. □

Proposition 2.3 *Let q be odd and let $c \in \mathbb{F}_q$ satisfy $c^2 = -1$. Then the vectors*

$$e_1 = (1, c), \quad e_2 = (2^{-1}, -2^{-1}c)$$

form a hyperbolic basis of $\mathbb{F}_q^2$. Equivalently, the line spanned by $(1, c)$ is isotropic and the Euclidean plane is split.

Proof. Direct calculation gives

$$\langle e_1, e_1 \rangle = 1 + c^2 = 0, \quad \langle e_2, e_2 \rangle = 2^{-2}(1 + c^2) = 0, \quad \langle e_1, e_2 \rangle = 2^{-1}(1 - c^2) = 1.$$

If $\alpha e_1 + \beta e_2 = 0$, pairing successively with e_2 and e_1 gives $\alpha = \beta = 0$. Hence e_1, e_2 form a hyperbolic basis. In particular, $\mathbb{F}_q(1, c)$ is isotropic and the Euclidean plane is split. □

Listing 4: Lean for Proposition 2.2 and Proposition 2.3.

```

theorem paper_split_consequences_exact
  [Fact ((2 : K) ≠ 0)] (c : K) (hc : c ^ 2 = (-1 : K)) :
  orderOf c = 4 ∧
  ∀( k : Nat,
    (c • (1 : Matrix (Fin k) (Fin k) K)) *
      (c • (1 : Matrix (Fin k) (Fin k) K)).transpose =
        -(1 : Matrix (Fin k) (Fin k) K)) ∧
  ∀( a : K, ∃ x y : K,
    x - c * y = 1 ∧ x + c * y = a ∧ x ^ 2 + y ^ 2 = a)

theorem paper_euclidean_plane_hyperbolic_basis_exact
  [Fact ((2 : K) ≠ 0)] (c : K) (hc : c ^ 2 = (-1 : K)) :
  paperHyperbolicPair (splitE1 c) (splitE2 c) ∧
  LinearIndependent K ![splitE1 c, splitE2 c]

```

3 Building-Up Theory

3.1 Binary Case

In 2001, Kim [8] proposed a building-up construction of binary self-dual codes in order to construct many self-dual codes of larger lengths from a self-dual code of a given length. An explicit form of the building-up construction is given below.

Theorem 3.1 ([8, Theorem 1]) *Let G_0 be a generator matrix of a binary self-dual code $\mathcal{C}_0$ of length $2n$, whose rows are $g_1, \dots, g_n$. Let $\mathbf{x}$ be a binary vector in $\mathbb{F}_2^{2n}$ satisfying $\mathbf{x} \cdot \mathbf{x} = 1$. Suppose that $y_i = \mathbf{x} \cdot g_i$ for $1 \leq i \leq n$. Then the following matrix*

$$G = \left[\begin{array}{cc|c} 1 & 0 & \mathbf{x} \\ \hline y_1 & y_1 & g_1 \\ \vdots & \vdots & \vdots \\ y_n & y_n & g_n \end{array} \right]$$

generates a binary self-dual code $\mathcal{C}$ of length $2n + 2$.

Then the converse of Theorem 3.1 was proved in [8] as follows.

Theorem 3.2 ([8, Theorem 2]) *Any binary self-dual code $\mathcal{C}$ of length $2n$ with minimum distance $d > 2$ is obtained from some self-dual code $\mathcal{C}_0$ of length $2n - 2$ (up to permutation equivalence) by the construction in Theorem 3.1.*

In 2012, Chinburg and Zhang [4] proved that, up to permutation equivalence, all binary self-dual codes of length at least four arise from Hilbert pairings on rings of S -integers of $\mathbb{Q}$.

Theorem 3.3 ([4, Theorem 1.5]) *Up to permutation equivalence, all binary self-dual codes of length at least four arise from Hilbert pairings on rings of S -integers of $K = \mathbb{Q}$. In fact, each such code arises up to permutation equivalence from infinitely many different subsets S of the places of $K = \mathbb{Q}$.*

Here equivalence has the precise meaning used by Chinburg–Zhang: a bijection of the selected orthonormal bases whose linear extension carries one code to the other. Thus it is permutation equivalence, not general monomial equivalence.

In the course of their proof, Chinburg–Zhang exhibit a boxed normal form for the resulting binary self-dual code. Its bottom row is the all-ones vector, its diagonal two-coordinate blocks are 01, and each remaining off-diagonal block is 00 or 11, with opposite blocks summing to 11. In the table, $W_v = \mathbb{Q}_v^\times / (\mathbb{Q}_v^\times)^2$ is the local square-class space with its Hilbert pairing; the column headings list local bases and the rows list global S -units.

S -units $\setminus$ places	W_{p_1} $\{-p_1, p_1\}$	W_{p_2} $\{-p_2, p_2\}$	$\dots$ $\dots$	W_2 $\{-2, -10, -5\}$	$W_{\mathbb{R}}$ $\{-1\}$
p_1	01	00/11		00/11	1 0
p_2	11/00	01		00/11	1 0
$\vdots$	$\vdots$		$\ddots$		$\vdots$
2	11/00	11/00		01	1 0
-1	11	11	$\dots$	11	1 1

The boxed reduction deletes one two-coordinate block and the corresponding pivot row. The comparison with Kim's construction must distinguish two bilinear forms. Let K be a number field, let S contain the archimedean and dyadic places, and put

$$U = \operatorname{Spec} \mathcal{O}_{K,S}, \quad W = \bigoplus_{v \in S} H_{\text{ét}}^1(K_v, \mu_2), \quad L = \operatorname{im} \left(H_{\text{ét}}^1(U, \mu_2) \longrightarrow W \right).$$

At real places the local groups use Tate cohomology. The sum of the local Hilbert pairings defines a nondegenerate form B_{coh} on W , and Chinburg–Zhang prove $L = L^{\perp B_{\text{coh}}}$. Under their local hypotheses, selected orthonormal bases give a form isometry

$$\iota_E : (W, B_{\text{coh}}) \xrightarrow{\sim} (\mathbb{F}_2^{2n}, \langle \cdot, \cdot \rangle_E).$$

Only after fixing ι_E do we regard the arithmetic image as a Euclidean binary code.

Theorem 3.4 (Euclidean realization and binary boxed reduction) *Let $n \geq 2$, let $L \subset W$ be the Chinburg–Zhang arithmetic image, and suppose that $L = L^{\perp B_{\text{coh}}}$. Fix the form isometry ι_E above and put $\mathcal{C} = \iota_E(L)$. Suppose that the coordinate permutation furnishing the Chinburg–Zhang boxed representative, followed by the indicated elementary row operation, gives*

$$G' = \left[\begin{array}{cc|c} 1 & 0 & x \\ \hline y_1 & y_1 & g_1 \\ \vdots & \vdots & \vdots \\ y_{n-1} & y_{n-1} & g_{n-1} \end{array} \right], \quad x, g_i \in \mathbb{F}_2^{2n-2}.$$

Let G have rows $g_1, \dots, g_{n-1}$, and put $\mathcal{C}' = \langle G \rangle_{\mathbb{F}_2}$. Then:

1. L is Lagrangian for B_{coh} , while $\mathcal{C}$ and the boxed representative $\langle G' \rangle_{\mathbb{F}_2}$ are Euclidean self-dual.
2. In the displayed Euclidean coordinates,

$$E_2 = \langle (1, 0), (0, 1) \rangle = \mathbb{F}_2^2, \quad \mathbb{F}_2^{2n} = E_2 \perp \mathbb{F}_2^{2n-2}, \quad \operatorname{Gram}(E_2) = I_2.$$

The plane E_2 is non-alternating and is not isometric to the alternating hyperbolic plane. Nevertheless, $\mathbb{F}_2(1, 1) \subset E_2$ is isotropic.

3. The rows g_i are linearly independent and pairwise orthogonal; hence $\mathcal{C}'$ is Euclidean self-dual of length $2n - 2$.
4. One has $x \cdot x = 1$ and, for every i ,

$$y_i = \langle x, g_i \rangle_E.$$

Consequently, G' is exactly the Kim building-up matrix obtained from G and x .

5. Deleting the top row and the first two coordinates of G' gives G , and rebuilding from G gives G' exactly. After restoring the coordinate permutation, the Chinburg–Zhang reduction and Kim extension are inverse up to permutation equivalence.

Proof. Let P_σ denote the selected permutation matrix and define $\eta(u) = \iota_E(u)P_\sigma$. Since right multiplication by P_σ and ι_E are form isometries,

$$\eta(u) \cdot \eta(v) = B_{\text{coh}}(u, v) \quad (u, v \in W).$$

Thus $\mathcal{C}_\sigma := \eta(L) = \langle G' \rangle_{\mathbb{F}_2}$ is Euclidean self-dual and has dimension n . Since ι_E is also a form isometry,

$$\mathcal{C} = \mathcal{C}^\perp, \quad \mathcal{C}_\sigma = \mathcal{C}_\sigma^\perp.$$

This proves (1).

The first two standard coordinate vectors $(1, 0)$ and $(0, 1)$ span the plane E_2 with Gram matrix I_2 . In particular,

$$(1, 0) \cdot (1, 0) = 1, \quad (1, 1) \cdot (1, 1) = 0, \quad (1, 0) \cdot (1, 1) = 1.$$

Hence E_2 is non-alternating and contains the isotropic line $\mathbb{F}_2(1, 1)$. The form with Gram matrix $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is alternating. Since alternation is invariant under isometry, the two planes are not isometric. This proves (2).

The rows of G' are independent and pairwise orthogonal. Writing $r_0 = (1, 0, x)$ and $r_i = (y_i, y_i, g_i)$, we obtain

$$0 = r_0 \cdot r_0 = 1 + x \cdot x, \quad 0 = r_0 \cdot r_i = y_i + x \cdot g_i,$$

and therefore

$$x \cdot x = 1, \quad y_i = x \cdot g_i.$$

Moreover, for all i, j ,

$$0 = r_i \cdot r_j = 2y_i y_j + g_i \cdot g_j = g_i \cdot g_j.$$

If $\sum_i \lambda_i g_i = 0$, then

$$\sum_i \lambda_i y_i = x \cdot \sum_i \lambda_i g_i = 0,$$

so $\sum_i \lambda_i r_i = 0$. Independence of the rows of G' gives $\lambda_i = 0$ for all i . Thus the g_i are independent and pairwise orthogonal. Since their number is $n - 1 = (2n - 2)/2$, their row space is self-dual. This proves (3), and the coefficient identities prove (4).

It remains to verify the inverse relation in (5). The first row of the Kim matrix constructed from x and G is $(1, 0, x) = r_0$, and its row corresponding to g_i is

$$(x \cdot g_i, x \cdot g_i, g_i) = (y_i, y_i, g_i) = r_i.$$

Hence the constructed matrix is equal to G' row by row. Conversely, deleting r_0 and the first two coordinates sends every remaining row r_i to g_i , and therefore recovers G . Thus reduction followed by building-up gives G' exactly. Since $\langle G' \rangle_{\mathbb{F}_2} = \mathcal{C}P_\sigma$, right multiplication by P_σ^{-1} shows that the reconstructed code is permutation equivalent to $\mathcal{C}$, as claimed. $\square$

Remark 3.5 The form isometry ι_E and the coordinate permutation P_σ play distinct roles. The former transports B_{coh} to the Euclidean dot product; the latter preserves that dot product. Neither identifies the deleted Euclidean plane with the alternating hyperbolic plane.

In the following listing, `e` denotes the composite $\eta(u) = \iota_E(u)P_\sigma$, so `hcode` identifies the actual permuted image of L with the displayed boxed family. The two plane lemmas for the standard Gram matrix and the characteristic-two non-isometry formalize item (2), while `paper_binary_cz_kim_corrected` formalizes the remaining reduction and reconstruction statements.

Listing 5: Lean for Theorem 3.4.

```
theorem paper_binary_cz_kim_corrected [CharP K 2]
  (he : IsFormIsometry Bcoh dotBilin e)
  (hL : IsSelfOrthogonal Bcoh L)
  (hcode : L.map e.toLinearMap =
    rowSpace (boxedFamily x Y G)) :
  IsSelfOrthogonal dotBilin (L.map e.toLinearMap) ∧
  dot x x = (1 : K) ∧
  paperSelfDualCode (rowSpace G) ∧
  ∀( i, Y i = dot x (G i)) ∧
  boxedFamily x Y G = buildRowsBin x G ∧
  deleteBinaryHeadPair (boxedFamily x Y G) = G ∧
  CodeEquiv
    (buildRowsBin x (deleteBinaryHeadPair (boxedFamily x Y G)))
    (boxedFamily x Y G)
```

3.2 Split q -ary Case

Motivated by the building-up construction of binary self-dual codes, Kim and Lee [10] generalized it to finite fields with q elements where $q \equiv 1 \pmod{4}$ as follows.

Theorem 3.6 (*[10, Proposition 2.1]*) *Let $\mathbb{F}_q$ be a finite field with q elements. Assume that $q \equiv 1 \pmod{4}$. Let c be in $\mathbb{F}_q$ such that $c^2 = -1$. Let G_0 be a generator matrix of a Euclidean self-dual code $\mathcal{C}_0$ over $\mathbb{F}_q$ of length $2n$, whose rows are $g_1, \dots, g_n$. Let $\mathbf{x}$ be a vector in $\mathbb{F}_q^{2n}$ satisfying $\mathbf{x} \cdot \mathbf{x} = -1$. Suppose that $y_i = \mathbf{x} \cdot g_i$ for $1 \leq i \leq n$. Then the following matrix*

$$G = \left[\begin{array}{cc|c} 1 & 0 & \mathbf{x} \\ \hline -y_1 & cy_1 & g_1 \\ \vdots & \vdots & \vdots \\ -y_n & cy_n & g_n \end{array} \right]$$

generates a self-dual code $\mathcal{C}$ over $\mathbb{F}_q$ of length $2n + 2$.

Remark 3.7 Replacing c by $-c$ preserves $c^2 = -1$. Hence we use the equivalent normalization $(-y_i, -cy_i, g_i)$ in the sequel.

Then the converse of the above theorem was also stated and proved in [10].

Theorem 3.8 ([10, Proposition 2.2]) *Let $\mathbb{F}_q$ be a finite field with q elements. Assume that $q \equiv 1 \pmod{4}$. Any Euclidean self-dual code $\mathcal{C}$ over $\mathbb{F}_q$ of length $2n$ with minimum distance $d > 2$ is obtained from some Euclidean self-dual code $\mathcal{C}_0$ over $\mathbb{F}_q$ of length $2n - 2$ (up to permutation equivalence) by the construction method in Theorem 3.6.*

This naturally leads to the following question: is there a split q -ary analogue of the Chinburg–Zhang boxed reduction for self-dual codes over $\mathbb{F}_q$ when $q \equiv 1 \pmod{4}$?

The next theorem gives the adapted split normal form that underlies the q -ary forward and reverse calculations. It identifies the hyperbolic decomposition, the shorter parent code, and the Kim–Lee correction formula once the self-dual code is presented in the displayed split form.

Theorem 3.9 (Adapted split reduction and building-up extension over $\mathbb{F}_q$) *Let q be a prime power with $q \equiv 1 \pmod{4}$, and let $K = \mathbb{F}_q$. Fix $c \in K^\times$ such that $c^2 = -1$. Let $W = K^{2n}$ be equipped with the standard Euclidean inner product*

$$\langle u, v \rangle = \sum_{i=1}^{2n} u_i v_i.$$

Let $L \subset W$ be a self-dual code, i.e., $L = L^\perp$. Assume that, after a permutation of coordinates and row operations, L admits a generator matrix of the form

$$M = \left[\begin{array}{cc|c} 1 & 0 & x \\ \hline -y_1 & -cy_1 & g_1 \\ \vdots & \vdots & \vdots \\ -y_{n-1} & -cy_{n-1} & g_{n-1} \end{array} \right],$$

where $x \in K^{2n-2}$ and $g_i \in K^{2n-2}$ for $1 \leq i \leq n - 1$.

Let $W_0 = K^{2n-2}$ and define $L_0 := \langle g_1, \dots, g_{n-1} \rangle \subset W_0$. Then the following statements hold:

1. *Let $E = K^2$ denote the first coordinate plane in the displayed decomposition of W . Then $W = E \perp W_0$ in the displayed coordinates, and E is isometric to a hyperbolic plane. This is an abstract change of basis in E ; no coordinate change is applied to the displayed matrix M .*

2. *L_0 is self-dual in W_0 .*

3. *One has*

$$\langle x, x \rangle = -1, \quad y_i = \langle x, g_i \rangle \quad (1 \leq i \leq n - 1).$$

4. *Consequently,*

$$L = \left\langle (1, 0, x), (-\langle x, g_i \rangle, -c\langle x, g_i \rangle, g_i) : 1 \leq i \leq n - 1 \right\rangle_K,$$

so that L is obtained from L_0 by a q -ary building-up construction.

5. Conversely, two reverse statements hold:

(5a) Let $L_0 \subset K^{2n-2}$ be self-dual, let $g_1, \dots, g_{n-1}$ be a basis of L_0 , and let $x \in K^{2n-2}$ satisfy

$$\langle x, x \rangle = -1,$$

then the row family generated by

$$r_0 = (1, 0, x), \quad r_i = (-\langle x, g_i \rangle, -c\langle x, g_i \rangle, g_i)$$

is linearly independent and generates a self-dual code of length $2n$.

(5b) More generally, if one starts from any generating family of a self-dual parent, the same formula always produces a self-orthogonal child family, and that child is self-dual as soon as its row span has dimension $n = \frac{1}{2} \dim W$.

Proof. The decomposition $W = E \perp W_0$, with $E = K^2$, is the literal decomposition into the first two and the remaining coordinates. The standard basis of E has Gram matrix I_2 . Since $2 \neq 0$, the vectors

$$h_1 = (1, c), \quad h_2 = \frac{1}{2}(1, -c)$$

satisfy $\langle h_1, h_1 \rangle = \langle h_2, h_2 \rangle = 0$ and $\langle h_1, h_2 \rangle = 1$. Thus E is abstractly hyperbolic, proving (1), while M remains written in the standard coordinate basis.

Write

$$r_0 = (1, 0, x), \quad r_i = (-y_i, -cy_i, g_i).$$

Self-duality of L gives pairwise orthogonality of these rows. Hence

$$0 = \langle r_0, r_0 \rangle = 1 + \langle x, x \rangle, \quad 0 = \langle r_0, r_i \rangle = -y_i + \langle x, g_i \rangle,$$

and

$$0 = \langle r_i, r_j \rangle = (1 + c^2)y_i y_j + \langle g_i, g_j \rangle.$$

Since $c^2 = -1$, it follows that

$$\langle x, x \rangle = -1, \quad y_i = \langle x, g_i \rangle, \quad \langle g_i, g_j \rangle = 0 \quad (1 \leq i, j \leq n-1).$$

The code L has dimension n , and the displayed n rows generate it. They are therefore linearly independent. Suppose that $\sum_i \lambda_i g_i = 0$. The coefficient identity just proved gives

$$\sum_i \lambda_i y_i = \left\langle x, \sum_i \lambda_i g_i \right\rangle = 0,$$

and therefore

$$\sum_i \lambda_i r_i = \left(-\sum_i \lambda_i y_i, -c \sum_i \lambda_i y_i, \sum_i \lambda_i g_i \right) = 0.$$

The independence of the rows of M implies $\lambda_i = 0$ for every i . Thus $g_1, \dots, g_{n-1}$ are linearly independent. Their span L_0 is self-orthogonal and has dimension $n-1 = \frac{1}{2} \dim W_0$, so

$L_0 = L_0^\perp$. This proves (2) and (3), and substitution of the coefficient identity into the rows proves (4).

Conversely, let L_0 be self-dual, put $y_i = \langle x, g_i \rangle$, and define r_0, r_i as above. Directly,

$$\langle r_0, r_0 \rangle = 1 + \langle x, x \rangle = 0, \quad \langle r_0, r_i \rangle = -y_i + \langle x, g_i \rangle = 0,$$

and

$$\langle r_i, r_j \rangle = (1 + c^2)y_i y_j + \langle g_i, g_j \rangle = 0.$$

Thus the child row space is self-orthogonal for every generating family. If $g_1, \dots, g_{n-1}$ is a basis, projection onto the last $2n - 2$ coordinates proves that $r_1, \dots, r_{n-1}$ are independent. Moreover, every linear combination of these successor rows has its first two coordinates in $K(1, c)$, whereas $(1, 0) \notin K(1, c)$. Hence r_0 is not in their span. The child therefore has dimension n and is self-dual. For an arbitrary generating family, the same Gram calculation and the stated half-dimension hypothesis give self-duality. $\square$

The following Lean theorem is the exact reverse statement used above. In particular, tail independence and its finrank conclusion precede the self-duality conclusion; no comparison of vectors in different ambient spaces occurs.

Listing 6: Lean for Theorem 3.9.

```
theorem paper_qary_adapted_reduction
  (hc : c ^ 2 = (-1 : K))
  (hself : paperSelfDualCode
    (rowSpace (qaryAdaptedFamily x c Y G))) :
  dot x x = (-1 : K) ∧
  ∀( i, Y i = dot x (G i)) ∧
  LinearIndependent K (qaryAdaptedFamily x c Y G) ∧
  LinearIndependent K G ∧
  Module.finrank K (rowSpace G) = m ∧
  paperSelfDualCode (rowSpace G) ∧
  qaryAdaptedFamily x c Y G = buildRows x c G
```

Remark 3.10 For a basis $g_1, \dots, g_{n-1}$ of the parent, the projection onto the last $2n - 2$ coordinates proves independence of the successor rows. For an arbitrary generating family, the same Gram calculation gives only self-orthogonality; the half-dimension hypothesis in Theorem 3.9(5b) is then necessary.

Remark 3.11 (Relation with Chinburg–Zhang) Theorem 3.9 is a statement about the Euclidean form over $\mathbb{F}_q$; it does not assert a μ_2 -cohomological realization. Its analogy with Theorem 3.4 is the common two-coordinate reduction governed by an isotropic line.

Theorem 3.9 is therefore the one-step adapted split theorem. It identifies the shorter parent code and recovers the Kim–Lee correction coefficients from the displayed hyperbolic

head. The next theorem packages the same mechanism into a boxed family, designed to play for $q \equiv 1 \pmod{4}$ the same structural role that boxed matrices play in the binary theorem of Chinburg–Zhang. The key difference is that in odd characteristic the last block of a non-final row can no longer be frozen to 10; instead it must vary through norm -1 vectors in the distinguished split 2-plane, with the last row and the off-diagonal isotropic blocks adjusted accordingly.

Theorem 3.12 (Split boxed form over $\mathbb{F}_q$) *Fix $c \in K^\times$ with $c^2 = -1$. For $1 \leq i \leq n-1$, let $\ell_i = (u_i, v_i) \in K^2$ and $a_i \in K$, and for distinct $1 \leq i, j \leq n-1$, let $b_{ij} \in K$. Let M be the $n \times 2n$ generator matrix whose $n \times n$ block form $\widetilde{M} = (B_{ij})$ of 1×2 blocks is determined by*

$$B_{ii} = (0, 1) \quad (1 \leq i \leq n-1), \quad B_{nn} = (1, c),$$

$$B_{in} = \ell_i \quad (1 \leq i \leq n-1), \quad B_{ni} = a_i(1, c) \quad (1 \leq i \leq n-1),$$

and

$$B_{ij} = b_{ij}(1, c) \quad (1 \leq i \neq j \leq n-1).$$

Thus $\widetilde{M}$ has the split boxed shape

$$\widetilde{M} = \left[\begin{array}{c|ccccc|c} 01 & b_{12}(1, c) & b_{13}(1, c) & \cdots & b_{1,n-1}(1, c) & \ell_1 \\ \hline b_{21}(1, c) & 01 & b_{23}(1, c) & \cdots & b_{2,n-1}(1, c) & \ell_2 \\ b_{31}(1, c) & b_{32}(1, c) & 01 & \cdots & b_{3,n-1}(1, c) & \ell_3 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ b_{n-1,1}(1, c) & b_{n-1,2}(1, c) & b_{n-1,3}(1, c) & \cdots & 01 & \ell_{n-1} \\ \hline a_1(1, c) & a_2(1, c) & a_3(1, c) & \cdots & a_{n-1}(1, c) & (1, c) \end{array} \right],$$

and the corresponding ordinary generator matrix M is

$$M = \left[\begin{array}{c|cc|cc|cc|cc|c} 0 & 1 & b_{12} & cb_{12} & b_{13} & cb_{13} & b_{14} & cb_{14} & \cdots & b_{1,n-1} & cb_{1,n-1} & \ell_1 \\ \hline b_{21} & cb_{21} & 0 & 1 & b_{23} & cb_{23} & b_{24} & cb_{24} & \cdots & b_{2,n-1} & cb_{2,n-1} & \ell_2 \\ \hline b_{31} & cb_{31} & b_{32} & cb_{32} & 0 & 1 & b_{34} & cb_{34} & \cdots & b_{3,n-1} & cb_{3,n-1} & \ell_3 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \vdots \\ \hline b_{n-1,1} & cb_{n-1,1} & b_{n-1,2} & cb_{n-1,2} & b_{n-1,3} & cb_{n-1,3} & b_{n-1,4} & cb_{n-1,4} & \cdots & 0 & 1 & \ell_{n-1} \\ \hline a_1 & ca_1 & a_2 & ca_2 & a_3 & ca_3 & a_4 & ca_4 & \cdots & a_{n-1} & ca_{n-1} & 1 \ c \end{array} \right],$$

where the final block column is still written in the shorthand $\ell_i = (u_i, v_i)$. Assume that

$$u_i^2 + v_i^2 = -1 \quad (1 \leq i \leq n-1), \quad (1)$$

$$u_i + cv_i = -ca_i \quad (1 \leq i \leq n-1), \quad (2)$$

and

$$c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0 \quad (1 \leq i < j \leq n-1), \quad (3)$$

where $\ell_i \cdot \ell_j = u_i u_j + v_i v_j$ is the Euclidean dot product on the distinguished 2-plane. Then the rows of M are pairwise orthogonal and linearly independent. Therefore M generates a self-dual code of length $2n$.

Proof. Let $R_1, \dots, R_{n-1}, R_n$ be the displayed rows, with R_n the last row. Since $(1, c) \cdot (1, c) = 0$, every block of R_n is isotropic, and therefore $R_n \cdot R_n = 0$. For $i < n$, all non-final blocks of R_i , except its diagonal block, lie on $K(1, c)$. Hence

$$R_i \cdot R_i = 1 + \ell_i \cdot \ell_i = 0$$

by (1), and

$$R_i \cdot R_n = (0, 1) \cdot a_i(1, c) + \ell_i \cdot (1, c) = ca_i + u_i + cv_i = 0$$

by (2). If $i < j < n$, all mixed block products vanish except those in block columns i, j, n . Thus

$$\begin{aligned} R_i \cdot R_j &= (0, 1) \cdot b_{ji}(1, c) + b_{ij}(1, c) \cdot (0, 1) + \ell_i \cdot \ell_j \\ &= c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0 \end{aligned}$$

by (3). The rows are pairwise orthogonal.

Suppose

$$\sum_{i=1}^{n-1} \lambda_i R_i + \lambda_n R_n = 0.$$

Apply the functional $\beta(x, y) = y - cx$ to the j -th non-final block. Since $\beta(0, 1) = 1$ and $\beta(t(1, c)) = 0$, the resulting equation is $\lambda_j = 0$. This holds for every $j < n$. The first coordinate of the final block then gives $\lambda_n = 0$. Hence the n rows are linearly independent. Their span is an n -dimensional totally isotropic subspace of K^{2n} , and therefore equals its orthogonal complement. $\square$

The Lean statement below has exactly the same block data and hypotheses as the displayed theorem. Its independence proof uses `splitBoxedReadout`, which applies $\beta(x, y) = y - cx$ to the pivot blocks and then reads the remaining coefficient from the final block.

Listing 7: Lean for Theorem 3.12.

```
theorem paper_split_boxed_form_exact
  (c : K) (ell : Fin m -> SplitBlock K)
  (a : Fin m -> K) (b : Fin m -> Fin m -> K)
  (hc : c ^ 2 = (-1 : K))
  (hnorm : ∀ i, dot (ell i) (ell i) = (-1 : K))
  (hlast : ∀ i, ell i 0 + c * ell i 1 = -c * a i)
  (hoff : ∀ i j, i < j ->
    c * (b i j + b j i) + dot (ell i) (ell j) = 0) :
```



```

SplitBoxedPairwiseOrthogonal
  (splitBoxedRows c ell a b) ∧
LinearIndependent K (splitBoxedRows c ell a b) ∧
splitBoxedRowSpace (splitBoxedRows c ell a b) =
  splitBlockRowBilin.orthogonal
    (splitBoxedRowSpace (splitBoxedRows c ell a b))

```

Theorem 3.12 is the forward boxed statement. The remaining question is how much of this boxed structure is forced by self-duality in the reverse direction once a self-dual code is placed in a suitable adapted form.

Theorem 3.13 (Conditional boxed normalization from a distinguished isotropic-line row)
Let $\mathcal{C} \subset K^{2n}$ be a self-dual code, and fix $c \in K^\times$ with $c^2 = -1$. Assume that, after coordinate permutations preserving the 2-block decomposition and elementary row operations, $\mathcal{C}$ admits a generator matrix whose last row is

$$(a_1(1, c), a_2(1, c), \dots, a_{n-1}(1, c), (1, c)),$$

and whose remaining rows $R_1, \dots, R_{n-1}$ satisfy the following pivot-adapted conditions:

1. *the i -th block of R_i is 01;*
2. *the last block of R_i is $\ell_i = (u_i, v_i)$;*
3. *every other non-final block of R_i lies on the same c -isotropic line, so for each $j \neq i, n$ one has*

$$B_{ij} = b_{ij}(1, c)$$

for some scalar $b_{ij} \in K$.

Then this generator matrix is automatically a split boxed matrix of Theorem 3.12: the self-duality of $\mathcal{C}$ forces

$$u_i^2 + v_i^2 = -1, \quad u_i + cv_i = -ca_i, \quad c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0,$$

and hence $\mathcal{C}$ is generated by a matrix of the displayed split boxed type.

Proof. Write the adapted generator matrix in block form as

$$\widetilde{M} = (B_{ij})_{1 \leq i, j \leq n},$$

with $B_{ii} = 01$ for $1 \leq i \leq n-1$, $B_{in} = \ell_i$, $B_{ni} = a_i(1, c)$, and $B_{ij} = b_{ij}(1, c)$ for $i \neq j < n$. Since $\mathcal{C}$ is self-dual, its generator rows are pairwise orthogonal.

Applying self-orthogonality to a non-final row R_i gives

$$0 = \langle R_i, R_i \rangle = 1 + (u_i^2 + v_i^2),$$

because the diagonal block contributes 1, the final block contributes $u_i^2 + v_i^2$, and every other block is isotropic. Hence

$$u_i^2 + v_i^2 = -1.$$

Next, orthogonality of R_i with the distinguished last row gives

$$0 = \langle R_i, R_n \rangle = (0, 1) \cdot a_i(1, c) + \ell_i \cdot (1, c) = ca_i + (u_i + cv_i),$$

so

$$u_i + cv_i = -ca_i.$$

Finally, for $1 \leq i < j \leq n - 1$, orthogonality of R_i and R_j gives

$$\begin{aligned} 0 = \langle R_i, R_j \rangle &= (0, 1) \cdot b_{ji}(1, c) + b_{ij}(1, c) \cdot (0, 1) + \ell_i \cdot \ell_j \\ &= c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j, \end{aligned}$$

since every other block product is a scalar multiple of $(1, c) \cdot (1, c) = 0$. These are precisely the relations in Theorem 3.12. $\square$

In Lean, the reverse statement is packaged as an existence theorem: from a distinguished isotropic-line row and pairwise orthogonality, one obtains a split-isometric code equivalence to the boxed building-up form together with an orthogonal tail family.

Listing 8: Lean for Theorem 3.13.

```
theorem paper_conditional_split_boxed_normalization_core
  (oi : Fin (m + 1))
  (g : Fin n → K) (ha : a ≠ 0)
  (hc : c ^ 2 = (-1 : K))
  (hrow : R oi = prepend2 a (c * a) g)
  (hall : ∀ j : Fin (m + 1), HasIsotropicHead (K := K) c (R j))
  (horth : PairwiseOrthogonal (K := K) R) :
  ∃ x : Fin n → K, ∃ a' : Fin m → K, ∃ G : Fin m → Fin n → K,
    SplitIsometryCodeEquiv (K := K) c R
      (buildRows x c (pivotResidualTails x a' G)) ∧
    IsOrthogonalTailFamily (K := K) x (pivotResidualTails x a' G)
```

Remark 3.14 Theorem 3.12 and Theorem 3.13 are converse only within the stated adapted presentation. For a fixed pairing of the scalar coordinates, the intersection with $\bigoplus_j K(1, c)$ may have dimension greater than one; Theorem 3.15 gives the exact rank- r form for that pairing. Theorem 3.16 then proves that a pairing of rank one always exists after a scalar-coordinate permutation.

3.3 Universal normal form

The preceding results distinguish two questions. The first concerns a fixed ordered pairing of the $2n$ coordinates; its exact normal form has a terminal rank determined by the code. The second allows a permutation of the scalar coordinates, and hence a change of pairing, while preserving the Euclidean form and every Hamming weight.

Theorem 3.15 (Universal rank- r boxed normal form over $\mathbb{F}_q$) *Let $K = \mathbb{F}_q$, where $q \equiv 1 \pmod{4}$, fix $c \in K^\times$ with $c^2 = -1$, and fix an ordered pairing $K^{2n} = (K^2)^n$. For every Euclidean self-dual code $\mathcal{C} \subseteq K^{2n}$, put*

$$U_c = \bigoplus_{j=1}^n K(1, c), \quad r = \dim(\mathcal{C} \cap U_c), \quad k = n - r.$$

After a permutation of the n two-coordinate blocks and elementary row operations, $\mathcal{C}$ has a generator matrix

$$G_r(c; Q, \ell, D) = \left[\begin{array}{c|c} \left(Q_{ij}(1, c) + \delta_{ij}(0, 1) \right)_{k \times k} & \left(\ell_{it} \right)_{k \times r} \\ \hline \left(-(D \partial_c \ell_j)_s(1, c) \right)_{r \times k} & \left(D_{st}(1, c) \right)_{r \times r} \end{array} \right], \quad (4)$$

where $1 \leq i, j \leq k$, $1 \leq s, t \leq r$, $\ell_{it} = (u_{it}, v_{it}) \in K^2$,

$$(\partial_c \ell_i)_t = v_{it} - cu_{it}, \quad Q_{ii} = \frac{c}{2}(1 + \ell_i \cdot \ell_i),$$

$D \in M_r(K)$ is nonsingular, and the remaining coefficients satisfy

$$c(Q_{ij} + Q_{ji}) + \ell_i \cdot \ell_j = 0 \quad (i \neq j). \quad (5)$$

The last r rows span $\mathcal{C} \cap U_c$. Conversely, every matrix (4) with $\det D \neq 0$ and (5) has $k + r$ independent, pairwise orthogonal rows and generates a Euclidean self-dual code. Moreover, retaining any subset of the first k rows together with the corresponding block columns and all terminal rows and columns gives a self-dual code of the same form with the same D .

Proof. For $v = (v_1, \dots, v_n) \in (K^2)^n$, write each block uniquely as

$$v_j = (\alpha_j(v), c\alpha_j(v) + \beta_j(v)), \quad \beta_j(v) = v_{j,2} - cv_{j,1},$$

and define $\partial : \mathcal{C} \rightarrow K^n$ by $\partial(v) = (\beta_1(v), \dots, \beta_n(v))$. Since $\ker \partial = \mathcal{C} \cap U_c$, rank-nullity gives $\text{rank } \partial = k$. Choose k coordinates on which $\text{im } \partial$ projects isomorphically, permute them to the first k positions, and choose rows $g_1, \dots, g_k \in \mathcal{C}$ whose defects restrict there to the standard basis. Complete them by a basis $S_1, \dots, S_r$ of $\ker \partial$. Applying ∂ to a linear relation first determines the coefficients of the g_i ; independence of the S_s then shows that these n rows form a basis of $\mathcal{C}$.

Let Q_{ij} be the first coordinate of the j -th pivot block of g_i , and write its terminal block-row as ℓ_i . Thus its j -th pivot block is $Q_{ij}(1, c) + \delta_{ij}(0, 1)$. Since every S_s lies in $\ker \partial$, write its pivot and terminal blocks as $A_{sj}(1, c)$ and $D_{st}(1, c)$, respectively. The identity

$$(\alpha, c\alpha + \beta) \cdot (\gamma, c\gamma + \delta) = c(\alpha\delta + \beta\gamma) + \beta\delta$$

and the orthogonality of the chosen basis give

$$A_{si} + (D\partial_c \ell_i)_s = 0, \quad \delta_{ij} + c(Q_{ij} + Q_{ji}) + \ell_i \cdot \ell_j = 0.$$

The first equation gives the lower-left block in (4); the second gives (5), and its diagonal part gives the stated formula for Q_{ii} . It remains to prove that D is nonsingular. Let $\lambda = (\lambda_1, \dots, \lambda_r) \in K^r$ satisfy $\lambda D = 0$. For every i , the first identity yields

$$(\lambda A)_i = \sum_{s=1}^r \lambda_s A_{si} = - \sum_{t=1}^r (\partial_c \ell_i)_t \sum_{s=1}^r \lambda_s D_{st} = 0.$$

Consequently, every pivot and terminal block of $\sum_s \lambda_s S_s$ is zero. Since the S_s are linearly independent, $\lambda = 0$. Thus $\ker D = 0$, and hence $\det D \neq 0$.

Conversely, suppose that D is nonsingular and that (5) holds. Denote the first k rows of (4) by $p_1, \dots, p_k$, put $q_{it} = (\partial_c \ell_i)_t$, and retain the notation $S_1, \dots, S_r$ for the terminal rows. Since $(1, c) \cdot (1, c) = 0$, one has

$$S_s \cdot S_u = 0, \quad p_i \cdot S_s = c \left(A_{si} + \sum_{t=1}^r q_{it} D_{st} \right) = 0.$$

Moreover,

$$p_i \cdot p_j = \begin{cases} 1 + 2cQ_{ii} + \ell_i \cdot \ell_i = 0, & i = j, \\ c(Q_{ij} + Q_{ji}) + \ell_i \cdot \ell_j = 0, & i \neq j. \end{cases}$$

Thus the displayed rows are pairwise orthogonal.

Suppose now that

$$\sum_{i=1}^k \xi_i p_i + \sum_{s=1}^r \mu_s S_s = 0.$$

Applying $(x, y) \mapsto y - cx$ to the j -th pivot block gives $\xi_j = 0$, because the defect of that block is δ_{ij} in p_i and zero in every S_s . Hence $\xi_1 = \dots = \xi_k = 0$. The first coordinates of the terminal blocks then give

$$0 = \sum_{s=1}^r \mu_s D_{st} = (\mu D)_t \quad (1 \leq t \leq r).$$

Since D is nonsingular, $\mu = 0$. The $k+r$ rows are therefore linearly independent and pairwise orthogonal in $K^{2(k+r)}$; their row space is consequently self-dual. If $x = \sum_i \xi_i p_i + \sum_s \mu_s S_s$ belongs to U_c , then the defect of its j -th pivot block is ξ_j , so $\xi_j = 0$ for every j . Conversely, every linear combination of the S_s lies in U_c . Therefore, writing $\mathcal{C}_r := \langle G_r(c; Q, \ell, D) \rangle_K$,

$$\mathcal{C}_r \cap U_c = \langle S_1, \dots, S_r \rangle_K.$$

Finally, let $I \subseteq \{1, \dots, k\}$. Retaining the rows p_i with $i \in I$, their pivot block columns, and all terminal rows and block columns replaces Q by $Q|_{I \times I}$ and $(\ell_i)_{i=1}^k$ by $(\ell_i)_{i \in I}$, while leaving D unchanged. The two Gram identities above restrict to indices in I . Hence its rows

are pairwise orthogonal; the pivot-block defects determine the coefficients of the retained p_i , and the nonsingularity of D determines those of the S_s . Thus the restricted matrix has $|I| + r$ independent rows in $K^{2(|I|+r)}$ and generates a self-dual code. For $I = \emptyset$, this row space is

$$\{(z_1(1, c), \dots, z_r(1, c)) : z \in K^r\},$$

the full terminal isotropic space in K^{2r} , because D is nonsingular. This proves the final assertion. $\square$

Example 3.1 (A fixed rank-two pairing over $\text{GF}(5)$) Take $c = 2$, so $c^2 = -1$, and let $\mathcal{C}$ be generated by

$$G_0 = \begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 1 & 1 & 3 \\ 3 & 4 & 2 & 1 & 4 & 4 & 3 & 2 \\ 1 & 4 & 3 & 4 & 2 & 3 & 4 & 2 \\ 1 & 2 & 2 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}.$$

All calculations in this example are in $\text{GF}(5)$. One checks that $G_0 G_0^T = 0$, while the minor formed by the first four columns has determinant 4. Hence $\mathcal{C}$ has dimension 4 and is self-dual.

Pair adjacent coordinates and let G_0^σ be obtained by ordering the four two-coordinate blocks as $(2, 4, 1, 3)$. Writing $a_i = Q_{ii}$, the diagonal formula in (4) gives $a_1 = 4$ and $a_2 = 3$. We substitute these values and $c = 2$, but retain the displayed diagonal formula in the second coordinate: the two blocks are $(4, 2 \cdot 4 + 1)$ and $(3, 2 \cdot 3 + 1)$. The invertible row transformation

$$T = \begin{pmatrix} 4 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 \\ 1 & 3 & 0 & 1 \\ 4 & 0 & 3 & 4 \end{pmatrix}, \quad \det T = 3,$$

gives

$$G_{2,2} = T G_0^\sigma = \left[\begin{array}{c|c|c|c} (4, 2 \cdot 4 + 1) & 2(1, 2) & (2, 1) & (3, 3) \\ \hline 4(1, 2) & (3, 2 \cdot 3 + 1) & (2, 3) & (0, 2) \\ \hline (1, 2) & 4(1, 2) & (1, 2) & (1, 2) \\ \hline 4(1, 2) & 2(1, 2) & (1, 2) & 2(1, 2) \end{array} \right].$$

This is (4) with

$$\ell_1 = ((2, 1), (3, 3)), \quad \ell_2 = ((2, 3), (0, 2)), \quad Q_{12} = 2, \quad Q_{21} = 4,$$

and

$$D = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad \det D = 1.$$

Modulo five the diagonal pairs are $(4, 4)$ and $(3, 2)$, respectively. For a block (u, v) , recall that $\partial_2(u, v) = v - 2u$. Hence

$$\begin{aligned} \partial_2 \ell_1 &= \begin{pmatrix} 1 - 2 \cdot 2 \\ 3 - 2 \cdot 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \\ -D \partial_2 \ell_1 &= -\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = -\begin{pmatrix} 4 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 4 \end{pmatrix}. \end{aligned}$$

Similarly, $\partial_2 \ell_2 = (4, 2)^\top$ and $-D\partial_2 \ell_2 = (4, 2)^\top$. The off-diagonal relation (5) is satisfied. Applying the blockwise defect map gives

$$\partial G_{2,2} = \begin{pmatrix} 1 & 0 & 2 & 2 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

so

$$r = \dim(\mathcal{C} \cap U_2) = 4 - \text{rank}(\partial G_{2,2}) = 2.$$

The block permutation preserves U_2 ; consequently, whole-block permutations and row operations cannot reduce r for this fixed oriented pairing.

Deleting pivot rows together with their corresponding block columns gives

$$G_{1,2} = \left[\begin{array}{c|cc} (3, 2 \cdot 3 + 1) & (2, 3) & (0, 2) \\ \hline 4(1, 2) & (1, 2) & (1, 2) \\ 2(1, 2) & (1, 2) & 2(1, 2) \end{array} \right], \quad G_{0,2} = \begin{pmatrix} (1, 2) & (1, 2) \\ (1, 2) & 2(1, 2) \end{pmatrix}.$$

Thus Theorem 3.15 gives the exact self-dual chain

$$[8, 4] \longrightarrow [6, 3] \longrightarrow [4, 2],$$

with the same terminal matrix D at every level and $\langle G_{0,2} \rangle_K = K(1, 2) \oplus K(1, 2)$.

The integer r in Theorem 3.15 belongs to the fixed pairing and can exceed one. The next theorem strengthens this result: for every nonzero-length code, a scalar-coordinate permutation can be chosen so that the new pairing has $r = 1$.

Theorem 3.16 (Universal rank-one normal form) *Let $K = \mathbb{F}_q$, where $q \equiv 1 \pmod{4}$, fix $c \in K^\times$ with $c^2 = -1$, and let $n > 0$. Put $k = n - 1$ and*

$$U_c = \bigoplus_{j=1}^n K(1, c) \subset (K^2)^n.$$

For $\ell_i = (u_i, v_i) \in K^2$, define

$$\delta_i = v_i - cu_i, \quad a_i = \frac{c}{2}(1 + \ell_i \cdot \ell_i) \quad (1 \leq i \leq k).$$

Given scalars $b_{ij} \in K$, $i \neq j$, let

$$G_1(c; b, \ell) = \left[\begin{array}{c|ccc|c} (a_1, ca_1 + 1) & b_{12}(1, c) & \cdots & b_{1k}(1, c) & \ell_1 \\ \hline b_{21}(1, c) & (a_2, ca_2 + 1) & \cdots & b_{2k}(1, c) & \ell_2 \\ \hline \vdots & \vdots & \ddots & \vdots & \vdots \\ \hline b_{k1}(1, c) & b_{k2}(1, c) & \cdots & (a_k, ca_k + 1) & \ell_k \\ \hline -\delta_1(1, c) & -\delta_2(1, c) & \cdots & -\delta_k(1, c) & (1, c) \end{array} \right],$$

with the evident one-row interpretation $G_1 = (1, c)$ when $n = 1$. For a linear code $\mathcal{C} \subset K^{2n}$, the following are equivalent:

1. $\mathcal{C} = \mathcal{C}^\perp$;

2. there are a permutation $\pi \in S_{2n}$ and data b, ℓ satisfying

$$c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0 \quad (i \neq j) \quad (6)$$

such that

$$\mathcal{C}P_\pi = \langle G_1(c; b, \ell) \rangle_K.$$

In these coordinates,

$$\dim((\mathcal{C}P_\pi) \cap U_c) = 1.$$

Moreover, π preserves the complete weight enumerator, and hence every coefficient A_j .

Proof. For a full-row-rank generator of a self-dual code $E \subset (K^2)^m$, write the paired scalar columns as

$$B = (A_1 \mid B_1 \mid \cdots \mid A_m \mid B_m)$$

and put $d_j = B_j - cA_j$. The map $z \mapsto zB$ identifies K^m with E , and $zB \in U_c$ if and only if $z[d_1 \cdots d_m] = 0$. Rank-nullity therefore gives

$$\dim(E \cap U_c) = m - \text{rank}[d_1 \cdots d_m]. \quad (7)$$

We prove by induction on m that the scalar coordinates can be paired so that the defect matrix has rank $m - 1$. For $m = 1$, a self-dual line is generated by $(1, c)$ or $(1, -c)$; interchanging the two scalar coordinates in the latter case gives the required orientation.

Assume $m > 1$. Self-orthogonality excludes words of weight one. If E has minimum distance greater than two, Theorem 3.8 gives, after a coordinate permutation, a Kim–Lee presentation with a self-dual parent E_0 . Extend an inductively chosen pairing of E_0 by the new coordinate pair. Relative to the top row and the parent rows, the defect columns are

$$(-c, 0), \quad (t_j - cs_j, d_j) \quad (1 \leq j < m),$$

where the d_j are the parent defect columns. Subtracting suitable multiples of the first column from the remaining columns removes their top entries, whence

$$\text{rank}\Delta_c(E) = 1 + \text{rank}\Delta_c(E_0) = m - 1.$$

Suppose instead that E contains a word of weight two. After permuting coordinates and rescaling that word, it is $(1, \varepsilon c, 0, \dots, 0)$ for some $\varepsilon \in \{1, -1\}$. Replacing each remaining generator v by $v - v_1(1, \varepsilon c, 0, \dots, 0)$ kills its first coordinate; orthogonality to the weight-two word then kills its second coordinate. Consequently

$$E = K(1, \varepsilon c) \perp E_0,$$

where $E_0 \subset K^{2m-2}$ is self-dual. If necessary, interchange the two coordinates of the first summand, so that it is generated by $(1, c)$.

Choose an inductive rank-one pairing of E_0 . Its $m - 1$ defect columns have rank $m - 2$; hence there is an index i for which $\{d_j : j \neq i\}$ is a basis of their span. Write the two scalar columns of the i -th parent block as A_i, B_i , so $d_i = B_i - cA_i$. Pairing the first new scalar coordinate with A_i , the second with B_i , and leaving the other parent pairs unchanged gives the defect columns

$$(-c, A_i), \quad (1, B_i), \quad (0, d_j) \quad (j \neq i).$$

The first column together with the last $m - 2$ columns is independent. Moreover,

$$A_i + cB_i = c(B_i - cA_i) = cd_i$$

lies in $\langle d_j : j \neq i \rangle$, and

$$(0, A_i + cB_i) - (-c, A_i) = c(1, B_i).$$

Thus the second column lies in the span of the independent $m - 1$ columns, so the new defect matrix has rank exactly $m - 1$. This completes the induction. By (7), there is a scalar-coordinate permutation π_0 such that

$$\dim((\mathcal{C}P_{\pi_0}) \cap U_c) = 1.$$

Let s span $(\mathcal{C}P_{\pi_0}) \cap U_c$. Some block of s is nonzero; after permuting whole blocks and rescaling s , take its last block to be $(1, c)$. The restriction to $\mathcal{C}P_{\pi_0}$ of

$$\partial_c((x_j, y_j)_{j=1}^n) = (y_j - cx_j)_{j=1}^n$$

has kernel Ks and rank $n - 1$. Select $n - 1$ independent coordinate functionals of its image and choose a complementary basis $p_1, \dots, p_k$ in reduced form. Then the i -th pivot block of p_i is $(a_i, ca_i + 1)$, every other pivot block is $b_{ij}(1, c)$, and the last block is $\ell_i = (u_i, v_i)$. Put $\delta_i = v_i - cu_i$. If the i -th block of s is $\lambda_i(1, c)$, then

$$0 = p_i \cdot s = c\lambda_i + u_i + cv_i = c(\lambda_i + \delta_i),$$

so $\lambda_i = -\delta_i$. The equations $p_i \cdot p_i = 0$ and $p_i \cdot p_j = 0$ now give

$$a_i = \frac{c}{2}(1 + \ell_i \cdot \ell_i), \quad c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0.$$

Hence this basis is exactly the row family of $G_1(c; b, \ell)$.

Conversely, assume (6). Let $p_1, \dots, p_k, s$ be the displayed rows of $G_1(c; b, \ell)$. Since each non-pivot block of s is a multiple of $(1, c)$, the definitions of δ_i and a_i give

$$p_i \cdot s = -c\delta_i + u_i + cv_i = 0, \quad p_i \cdot p_i = 1 + 2ca_i + \ell_i \cdot \ell_i = 0,$$

and (6) gives

$$p_i \cdot p_j = c(b_{ij} + b_{ji}) + \ell_i \cdot \ell_j = 0.$$

Thus the rows are pairwise orthogonal. Applying ∂_c to the first k blocks of a relation among them returns the coefficients of $p_1, \dots, p_k$; all are zero, and the last block then forces the coefficient of s to vanish. The rows are therefore independent, and their n -dimensional totally isotropic span in K^{2n} is self-dual. Finally, coordinate permutations preserve the Euclidean form and every Hamming weight, hence the complete weight enumerator. $\square$

The fixed-pairing rank in Example 3.1 does not preclude a scalar-coordinate permutation that changes the pairing. The next example applies Theorem 3.16 to the same code and, at the same time, shows that rank one does not force the diagonal blocks to be $(0, 1)$.

Example 3.2 (The rank-one permutation of Example 3.1) Return to the original generator G_0 in Example 3.1. Let τ interchange only its first two scalar coordinates, so the ordered pairs become $(2, 1), (3, 4), (5, 6), (7, 8)$. With

$$T_1 = \begin{pmatrix} 0 & 2 & 2 & 0 \\ 2 & 2 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 1 & 4 & 1 & 1 \end{pmatrix}, \quad \det T_1 = 4,$$

exact row reduction gives

$$G_{3,1} = T_1 G_0^\tau = \left[\begin{array}{c|c|c|c} (1, 2 \cdot 1 + 1) & 0(1, 2) & 2(1, 2) & (4, 3) \\ \hline 4(1, 2) & (0, 2 \cdot 0 + 1) & 0(1, 2) & (3, 0) \\ \hline 1(1, 2) & 2(1, 2) & (4, 2 \cdot 4 + 1) & (2, 2) \\ \hline 0(1, 2) & 1(1, 2) & 2(1, 2) & (1, 2) \end{array} \right].$$

Thus its three diagonal blocks display the substituted pairs $(1, 2 \cdot 1 + 1)$, $(0, 2 \cdot 0 + 1)$, and $(4, 2 \cdot 4 + 1)$, corresponding to $(a_1, a_2, a_3) = (1, 0, 4)$ with $c = 2$. Its defect matrix is

$$\partial_2 G_{3,1} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 4 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad \dim((\mathcal{C}P_\tau) \cap U_2) = 4 - 3 = 1.$$

More precisely, $G_{3,1}$ is the matrix $G_1(c; b, \ell)$ of Theorem 3.16 with

$$\ell_1 = (4, 3), \quad \ell_2 = (3, 0), \quad \ell_3 = (2, 2),$$

$$(a_1, a_2, a_3) = (1, 0, 4), \quad (b_{12}, b_{13}, b_{21}, b_{23}, b_{31}, b_{32}) = (0, 2, 4, 0, 1, 2).$$

Indeed, $(\delta_1, \delta_2, \delta_3) = (0, 4, 3)$, so the last row has coefficients $-(\delta_1, \delta_2, \delta_3) = (0, 1, 2)$, and all three relations (6) hold. Thus the permutation does not merely lower a numerical defect rank: it produces the literal universal rank-one generator with $D = (1)$. The nonzero diagonal coefficients a_1 and a_3 show explicitly why Theorem 3.16 does not imply the zero-diagonal specialization of Theorem 3.12.

The formal proof separates fixed-pairing normalization, universal rank-one pairing, and the forward self-duality calculation:

Listing 9: Lean goals for Theorems 3.15 and 3.16.

```
theorem every_qary_selfDualCode_has_rankOne_orientedPairing
  (hn : 0 < n) (c : K) (hc : c ^ 2 = (-1 : K))
  (h2 : (2 : K) != 0)
```

```

(hC : QaryBlockSelfDualCode C) :
HasQaryRankOneOrientedPairing c C

theorem every_qary_selfDualCode_has_rankBoxed_normalForm
(c : K) (hc : c ^ 2 = (-1 : K))
(h2 : (2 : K) != 0)
(hC : QaryBlockSelfDualCode C) :
HasQaryRankBoxedNormalForm c C

theorem paperRankBoxedRows_forward_selfDual
(hc : c * c = -1) (h2 : (2 : K) != 0)
(hD : RankBoxCoreFullRank D)
(hoff : PaperOffDiagonalRelations c Q ell) :
RankBoxedPairwiseOrthogonal
  (paperRankBoxedRows c Q ell D) /\
LinearIndependent K (paperRankBoxedRows c Q ell D) /\
paperSelfDualCode
  (rankBoxedRowSpace (paperRankBoxedRows c Q ell D))

```

Remark 3.17 The split boxed form of Theorem 3.12 is the specialization $\ell_i \cdot \ell_i = -1$ of Theorem 3.16, since then $a_i = \frac{c}{2}(1 + \ell_i \cdot \ell_i) = 0$.

4 Applications

The applications are arranged around two complementary uses of the normal forms. The first consists of two optimal self-dual codes over $\text{GF}(5)$ linked by a boxed reduction and extension. The second is a single nontrivial $\text{GF}(13)$ realization in which the rank-one parent and its repeated building-up child are visible in the same complete generator matrix. Thus the section records mechanisms, rather than a list of unrelated parameter sets.

4.1 Optimal self-dual codes over $\text{GF}(5)$

Example 4.1 (An optimal self-dual $[6, 3, 4]$ code over $\text{GF}(5)$) Let $K = \text{GF}(5)$ and $c = 2$, so $c^2 = -1$. Choose

$$\ell_1 = (0, 2), \quad \ell_2 = (0, 2), \quad b_{12} = 1, \quad b_{21} = 2.$$

Then (2) gives

$$a_1 = a_2 = 3,$$

and the corresponding split boxed matrix is

$$\widetilde{M}_3 = \begin{pmatrix} 01 & 12 & 02 \\ 24 & 01 & 02 \\ 31 & 31 & 12 \end{pmatrix}.$$

This matrix generates an optimal self-dual $[6, 3, 4]$ code over $\text{GF}(5)$.

Proof. The boxed relations are satisfied, so Theorem 3.12 shows that $\widetilde{M}_3$ generates a self-dual $[6, 3]$ code. In ordinary coordinates the generator matrix is

$$G_3 = \left(\begin{array}{cc|cc|cc} 0 & 1 & 1 & 2 & 0 & 2 \\ 2 & 4 & 0 & 1 & 0 & 2 \\ 3 & 1 & 3 & 1 & 1 & 2 \end{array} \right).$$

Enumeration of the 5^3 coefficient vectors gives the weight enumerator

$$1 + 60z^4 + 24z^5 + 40z^6.$$

Hence the minimum distance is 4. The Singleton bound gives $d \leq 6 - 3 + 1 = 4$ for every linear $[6, 3]$ code, so this code is optimal.

Deleting the first block row and the first block column from $\widetilde{M}_3$ leaves the shorter parent package

$$\widetilde{M}_2 = \begin{pmatrix} 01 & 02 \\ 31 & 12 \end{pmatrix},$$

which is the corresponding self-dual $[4, 2, 2]$ parent. $\square$

Example 4.2 (A split boxed child $[8, 4, 4]$ over $\text{GF}(5)$) Let $K = \text{GF}(5)$ and $c = 2$. Choose

$$\ell_1 = \ell_2 = \ell_3 = (0, 2),$$

and

$$b_{12} = b_{13} = b_{23} = 1, \quad b_{21} = b_{31} = b_{32} = 2.$$

Then (2) gives

$$a_1 = a_2 = a_3 = 3,$$

and the corresponding split boxed matrix is

$$\widetilde{M}_4 = \begin{pmatrix} 01 & 12 & 12 & 02 \\ 24 & 01 & 12 & 02 \\ 24 & 24 & 01 & 02 \\ 31 & 31 & 31 & 12 \end{pmatrix}.$$

This matrix generates an optimal self-dual $[8, 4, 4]$ code over $\text{GF}(5)$.

Proof. Again the boxed relations hold, so Theorem 3.12 gives a self-dual $[8, 4]$ code. In ordinary coordinates the generator matrix is

$$G_4 = \left(\begin{array}{cc|cc|cc|cc} 0 & 1 & 1 & 2 & 1 & 2 & 0 & 2 \\ 2 & 4 & 0 & 1 & 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & 0 & 1 & 0 & 2 \\ 3 & 1 & 3 & 1 & 3 & 1 & 1 & 2 \end{array} \right).$$

Enumeration of the 5^4 coefficient vectors gives

$$1 + 48z^4 + 32z^5 + 288z^6 + 128z^7 + 128z^8,$$

so the minimum distance is 4. If a linear $[8, 4]$ code over $\text{GF}(5)$ had distance at least (5), the Singleton bound would force distance (5), making it a nontrivial MDS code of length $8 > q + 1 = 6$, which is impossible. Thus the displayed self-dual code is optimal. $\square$

Moreover, deleting the first block row and the first block column from $\widetilde{M}_4$ recovers exactly $\widetilde{M}_3$. Thus these two examples constitute one recursive calculation,

$$[4, 2, 2]_5 \longrightarrow [6, 3, 4]_5 \longrightarrow [8, 4, 4]_5,$$

rather than two unrelated existence examples.

4.2 A repeated boxed realization over $\text{GF}(13)$

Let $c = 5 \in \text{GF}(13)$, so $c^2 = -1$. If G has rows $g_1, \dots, g_m$, put

$$\mathcal{B}_5(G, x) = \left[\begin{array}{cc|c} 0 & 1 & x \\ \hline -5y_1 & -y_1 & g_1 \\ \vdots & \vdots & \vdots \\ -5y_m & -y_m & g_m \end{array} \right], \quad y_i = x \cdot g_i, \quad x \cdot x = -1. \quad (8)$$

This is the matrix in Theorem 3.9 after interchanging its first two coordinates. Thus the new diagonal block is $(0, 1)$, in the same orientation as the diagonal blocks of the rank-one parent.

Example 4.3 (A repeated boxed realization over $\text{GF}(13)$) Let $\mathcal{C}_{20}^{(13)}$ be the pure double-circulant self-dual code generated by

$$G_{20} = [I_{10} \mid \text{circ}(3, 7, 5, 5, 2, 8, 5, 6, 3, 0)].$$

It has parameters $[20, 10, 10]$ and $A_{10} = 111024$ [6]. After the ordered two-coordinate reduction at $(11, 7)$, followed by the recorded row normalization and coordinate permutation, one obtains a self-dual parent $\mathcal{C}_{18}^{(13)}$ with parameters $[18, 9, 8]$ and $A_8 = 1896$. There are a rank-one generator $G_{18}^{(13)}$ of this parent and

$$x_{18}^{(13)} = (2, 0, 0, 11, 0, 5, 0, 0, 6, 7, 6, 6, 0, 9, 0, 0, 0, 1)$$

such that

$$x_{18}^{(13)} \cdot x_{18}^{(13)} = -1, \quad \langle \mathcal{B}_5(G_{18}^{(13)}, x_{18}^{(13)}) \rangle_{\text{GF}(13)} = \mathcal{C}_{20}^{(13)}.$$

In $G_{18}^{(13)}$, every pivot diagonal block is $(0, 1)$, every off-diagonal block is $b_{ij}(1, 5)$, and the terminal block is $(1, 5)$. Hence the parent is literally in the zero-diagonal rank-one form of Theorem 3.12, while $\mathcal{B}_5(G_{18}^{(13)}, x_{18}^{(13)})$ is its Kim–Lee building-up child.

Proof. Exact calculation over $\text{GF}(13)$ gives

$$\text{rank} G_{20} = 10, \quad G_{20} G_{20}^T = 0,$$

and its complete weight distribution has first nonzero coefficient $A_{10} = 111024$. The two-coordinate reduction at $(11, 7)$ has rank 9, zero Gram matrix, and first nonzero weight coefficient $A_8 = 1896$; it therefore generates the asserted self-dual parent.

For the normalized parent, direct blockwise calculation verifies the diagonal, off-diagonal, and terminal relations stated above. Theorem 3.12 therefore gives its self-duality. Moreover, the displayed correction has norm -1 , and multiplication gives $y_i = x_{18}^{(13)} \cdot g_i$ for every parent row. Swapping the first two columns of (8) therefore gives exactly the forward matrix in Theorem 3.9. Its row space agrees with the row space of G_{20} after the recorded permutation, which proves the reconstruction identity. $\square$

Figure 1 displays the complete matrix, not only its block pattern. Blue marks the newly adjoined row, and orange marks the new coordinate pair in the lower rows. Inside the parent, teal marks the pivot blocks, violet marks the terminal row and terminal pair, and light gray marks the diagonal two-coordinate blocks. The outer thick rules separate the building-up border; the inner thick rules separate the pivot part of the rank-one parent from its terminal part.

$$\tilde{G}_{20}^{(13)} = \left(\begin{array}{c|c|cccccccccccccc} \text{rows} & \text{new pair} & & & & & & & & & & & & & & \\ \hline & & \text{diag: } 01; & i \neq j: b_{ij}(1,c) & & & & & & & & & & & D=(1): \ell_i; (1,c) \\ \hline 01|x & 0 & 1 & 2 & 0 & 0 & 11 & 0 & 5 & 0 & 0 & 6 & 7 & 6 & 6 & 0 & 9 & 0 & 0 & 0 & 1 \\ \hline b_{ij}, \ell_i & 2 & 3 & 0 & 1 & 10 & 11 & 12 & 8 & 7 & 9 & 9 & 6 & 0 & 0 & 6 & 4 & 7 & 9 & 9 & 3 \\ b_{ij}, \ell_i & 0 & 0 & 11 & 3 & 0 & 1 & 9 & 6 & 4 & 7 & 2 & 10 & 2 & 10 & 11 & 3 & 0 & 0 & 9 & 3 \\ b_{ij}, \ell_i & 2 & 3 & 5 & 12 & 8 & 1 & 0 & 1 & 9 & 6 & 10 & 11 & 6 & 4 & 10 & 11 & 12 & 8 & 5 & 0 \\ b_{ij}, \ell_i & 6 & 9 & 9 & 6 & 12 & 8 & 7 & 9 & 0 & 1 & 6 & 4 & 4 & 7 & 3 & 2 & 0 & 0 & 10 & 4 \\ b_{ij}, \ell_i & 7 & 4 & 1 & 5 & 8 & 1 & 0 & 0 & 12 & 8 & 0 & 1 & 1 & 5 & 8 & 1 & 11 & 3 & 3 & 9 \\ b_{ij}, \ell_i & 2 & 3 & 10 & 11 & 8 & 1 & 7 & 9 & 5 & 12 & 3 & 2 & 0 & 1 & 10 & 11 & 9 & 6 & 0 & 5 \\ b_{ij}, \ell_i & 12 & 5 & 3 & 2 & 11 & 3 & 8 & 1 & 7 & 9 & 8 & 1 & 3 & 2 & 0 & 1 & 6 & 4 & 8 & 0 \\ b_{ij}, \ell_i & 7 & 4 & 2 & 10 & 9 & 6 & 6 & 4 & 10 & 11 & 5 & 12 & 4 & 7 & 2 & 10 & 0 & 1 & 8 & 0 \\ \hline a_{i,(1,c)} & 3 & 11 & 3 & 2 & 3 & 2 & 12 & 8 & 7 & 9 & 6 & 4 & 8 & 1 & 1 & 5 & 1 & 5 & 1 & 5 \end{array} \right).$$

Figure 1: Repeated boxed realization $\mathcal{C}_{18}^{(13)} \longleftrightarrow \mathcal{C}_{20}^{(13)}$.

Thus the same figure exhibits both directions:

$$\mathcal{C}_{18}^{(13)} \xrightarrow{\mathcal{B}_5(G_{18}^{(13)}, x_{18}^{(13)})} \mathcal{C}_{20}^{(13)}, \quad \mathcal{C}_{20}^{(13)} \xrightarrow{(11,7)\text{-reduction}} \mathcal{C}_{18}^{(13)}.$$

Both distances equal the published best-known distances at their respective lengths [6, 9]. The complete audit of all $20 \cdot 19 = 380$ ordered coordinate pairs also identifies the limitation of this fixed child: every reduction loses a weight-ten word, and the smallest loss is 1896, attained by ten ordered pairs. Hence this code does not produce a self-dual $[18, 9, 9]$ parent by one two-coordinate reduction; this is a statement about the fixed code, not a nonexistence theorem.

Thus one bidirectional calculation simultaneously displays the universal rank-one parent, the repeated building-up border, exact reconstruction, and the behavior of the minimum-weight words.

5 Conclusion

We have separated the binary arithmetic comparison from the split odd-characteristic geometry, proved the adapted q -ary forward and reverse statements, and then removed the adapted-presentation hypothesis first by a universal rank- r normal form for every fixed pairing and then by a universal rank-one form after a scalar-coordinate permutation. Thus the same two-coordinate mechanism supports binary reduction, Kim–Lee extension, and a universal permutation-equivalent representative without identifying the binary Euclidean plane with an alternating hyperbolic plane. The applications show the mechanism first in a short quinary recursion and then in the complete repeated boxed realization $[18, 9, 8]_{13} \leftrightarrow [20, 10, 10]_{13}$, where the rank-one parent and its building-up child are visible in one generator matrix. Lean verifies the algebraic interfaces and their dependencies; the arithmetic realization theorem and finite distance enumerations remain explicitly identified mathematical and computational inputs.

Availability of the formalization

The modular development uses Lean v4.29.0-rc6 and Mathlib. Its nineteen independent Comparator suites check 181 declarations, including the exact universal pairing goal. Production proofs use neither `sorry` nor `native_decide`; their transitive axioms are confined to `propext`, `Quot.sound`, and `Classical.choice`. Reproducible finite-field calculations and the application verification records are included in the same public repository:

LeGenAI/intersection-coding-theory-cohomology.

References

- [1] C. Bachoc and P. Gaborit, “Designs and self-dual codes with long shadows,” *J. Combin. Theory Ser. A*, 2004, 105(1): 15-34.
- [2] E. Bannai, S. T. Dougherty, M. Harada, and M. Oura, Type II codes, “even unimodular lattices, and invariant rings,” *IEEE Trans. Inf. Theory*, 1999, 45(4): 1194-1205.
- [3] A. R. Calderbank, E. M. Rains, P. M. Shor, and N. J. A. Sloane, Quantum error correction via codes over GF(4), *IEEE Trans. Inf. Theory*, 1998, 44(4): 1369-1387.

- [4] T. Chinburg and Y. Zhang, “Every binary self-dual code arises from Hilbert symbols,” *Homology, Homotopy Appl.*, vol. 14, no. 2, pp. 189–196, 2012.
- [5] J. H. Conway, N. J. A. Sloane, Sphere Packings, Lattices and Groups, 3rd edn. Springer, New York (1998).
- [6] K. Betsumiya, S. Georgiou, T. A. Gulliver, M. Harada, and C. Koukouvinos, “On self-dual codes over some prime fields,” *Discrete Math.*, vol. 262, nos. 1–3, pp. 37–58, 2003.
- [7] W. C. Huffman and V. Pless, *Fundamentals of Error-Correcting Codes*, Cambridge University Press, 2003.
- [8] J.-L. Kim, “New extremal self-dual codes of lengths 36, 38, and 58,” *IEEE Trans. Inform. Theory*, vol. 47, no. 1, pp. 386–393, 2001.
- [9] J.-L. Kim and W.-H. Choi, “Self-dual codes, symmetric matrices, and eigenvectors,” *IEEE Access*, vol. 9, pp. 104294–104303, 2021.
- [10] J.-L. Kim and Y. Lee, “Euclidean and Hermitian self-dual MDS codes over large finite fields.” *J. Combin. Theory Ser. A* 105(1):79-95, 2004.
- [11] J.-L. Kim and Y. Lee, “Self-dual codes using the building-up construction,” *IEEE Int. Symp. Inform. Theory*, Seoul, Korea (South), 2009, pp. 2400–2402.
- [12] M. Kreck and V. Puppe, “Involutions on 3-manifolds and self-dual, binary codes,” *Homology, Homotopy Appl.*, vol. 10, no. 2, pp. 139–148, 2008.
- [13] J. S. Leon, V. Pless, and N. J. A. Sloane, “Self-dual codes over $\text{GF}(5)$,” *J. Combin. Theory Ser. A*, vol. 32, no. 2, pp. 178–194, 1982.
- [14] H. Montani, “Lagrangian and orthogonal splittings, quasitriangular Lie bialgebras and almost complex product structures,” *J. Math. Phys.*, vol. 64, 053505, 2023.
- [15] J. S. Milne, Arithmetic duality theorems, 2nd ed., BookSurge Publishing, Charleston, SC, 2006.
- [16] V. Pless, “The number of isotropic subspaces in a finite geometry,” *Atti Accad. Naz. Lincei Rend.*, vol. 39, pp. 418–421, 1965.
- [17] E. Rains and N. J. A. Sloane, Self-dual codes, in: V. S. Pless, W. C. Huffman (Eds.), *Handbook of Coding Theory*, Elsevier, Amsterdam. The Netherlands. (1998)
- [18] J.-P. Serre, A course in arithmetic, Grad. Texts in Math. 7, Springer-Verlag, New York, 1973.
- [19] C. Shannon, “Communication Theory of Secrecy Systems,” *Bell System Technical Journal*, vol. 28, no. 4, pp. 656–715, 1949.
- [20] M. Shi, Y. J. Choie, A. Sharma, and P. Solé, *Codes and Modular Forms: A Dictionary*. World Scientific, 2020.